

TUTORIAL QUESTIONS

CHAPTER 1 ORIENTATION TO CIVIL ENGINEERING

1. Where and when did engineering originate?
2. Give at least three examples of early engineering works.
3. Name the three main branches of engineering.
4. What does civil engineering deal with?
5. Why were the words “civil engineering” chosen to describe civil engineering?
6. When was civil engineering first taught at university level and in which country?
7. Explain why life would be primitive if engineering was not there.
8. Describe the four cluster subjects referred to as STEM.
9. What is the basis of civil engineering?
10. What is the definition of civil engineering?

CHAPTER 2 HISTORY OF CIVIL ENGINEERING

1. Name five historical periods before modern ages.
2. Name the five simple machines which were invented during the bronze age.
3. What is artisanship? When did artisanship start to develop?
4. Who is an architect-engineer? Who is a master builder?
5. Name the first architect-engineer known by name. Describe his work.
6. Name three iron age philosophers who contributed to science and mathematics.
7. State the key historical event during each of the 17th, 18th and 19th centuries.
8. Who is the father of modern science? What was his contribution to engineering?
9. Name four mathematicians who contributed to civil engineering in the 18th century.
10. Give a brief outline of civil engineering advances in the 20th century.

CHAPTER 3 AREAS OF SPECIALIZATION

1. Why is the civil engineering discipline sub-divided into sub-disciplines?
2. What is an area of specialization?
3. How can a civil engineer specialize?
4. Name five areas of specialization within civil engineering.
5. What does structural engineering specialize in?
6. What does geotechnical engineering specialize in?
7. What does transportation engineering specialize in?
8. What does water resources engineering specialize in?

CHAPTER 4 THE ENGINEER AS A PROFESSIONAL

PART 1

1. What is a profession?
2. Who is a professional?
3. What are professional ethics?
4. What is professionalism?
5. Outline the structure of the engineering profession.
6. State the four categories of engineers.
7. Name at least three bodies which regulate the engineering profession.
8. Describe the origins of professional ethics.
9. State nine behavioral qualities which are used define ethical behavior.
10. How are professional ethics enforced?

PART 2

1. State and explain the three characteristics which define structural behavior.
2. Distinguish between structural analysis and structural design.
3. Outline the basis of structural design.
4. What is the aim of structural design?
5. State the four requirements that structural design must satisfy.
6. Outline the eight steps used to accomplish structural design.

PART 3

1. Why must the engineer supervise the construction work?
2. Name the five parties involved in construction work.
3. What is maintenance?
4. Distinguish between routine maintenance and scheduled maintenance.
5. Distinguish between demolition and deconstruction.
6. Compare the strategy used in safe demolition with the strategy used in deconstruction.